

Decompose a Goal

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Follow the execution order for Functions and arguments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Functions and arguments. Proposed explanation: “Writing def runs the body immediately”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Create a new function independently

Each sealed box contains 6 counters. Define `show_stock(boxes, loose)` to print the total counters using its two parameters. Assume nonnegative integer arguments. Call it with (3, 2), (0, 2) and (3, 0). Predict each output and explain why `show_stock(2, 3)` is a different case.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Decompose a recommender for books under 120 pages in a requested genre. A teammate scores every book before checking the page limit. Explain a clearer order and the record passed between stages.

6. Solve the given case

A class wants to recommend one of five library books.

Decompose the goal into checkable steps.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A recommender must receive a topic, find matching books and show one title. Split the goal into three tasks and state what passes between them.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 2 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Separate reader profile, book data, score calculation, ranking, and explanation.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Decompose a recommender for books under 120 pages in a requested genre. A teammate scores every book before checking the page limit. Explain a clearer order and the record passed between stages.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Label the parts

In `def show_remaining(capacity, used)`, identify the function name and parameters. In `show_remaining(15, 6)`, identify the arguments and their matching parameters.

My reasoning and evidence

13. W2 Follow the execution order

Using the correct definition, predict `show_remaining(8, 3)`, `show_remaining(8, 8)` and `show_remaining(8, 0)` in that order. Explain what happens if the file contains only the definition.

My reasoning and evidence

14. W3 Repair ignored arguments

The faulty body calculates $12 - 4$ regardless of the call. State the output and expected output for `show_remaining(11, 2)`. Repair the expression and name an earlier call to rerun.

My reasoning and evidence

15. W4 Create a new function independently

Each sealed box contains 6 counters. Define `show_stock(boxes, loose)` to print the total counters using its two parameters. Assume nonnegative integer arguments. Call it with $(3, 2)$, $(0, 2)$ and $(3, 0)$. Predict each output and explain why `show_stock(2, 3)` is a different case.

My reasoning and evidence

16. W5 State the input assumption

The correct subtraction function prints -1 for `show_remaining(4, 5)`. Does successful execution make this a valid cupboard record? Explain what the function assumes and what needs checking.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record:

Evidence: First step I repaired and why

My record:

Evidence: New case checked later; date and result

My record:

Evidence: What I can now explain without notes

My record:

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Decompose a Goal

A fictional library computes total spine labels needed for several equal shelves.

Define `labels(shelves,per_shelf)`: return `shelves*per_shelf`.

Call `needed=labels(4,6)`, then `print(needed)`. No other side effects are defined.

AI ACADEMY / AI-2/2

Decompose a Goal

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Call with arguments

The call supplies `shelves=4` and `per_shelf=6` locally.



2. Function body

Multiply 4×6 to calculate 24.



3. Return value

`return` sends 24 back to the caller; it does not print.



4. Caller prints

`needed` receives 24; `print(needed)` displays 24.

CHANGE ONE THING

Call `labels(4,0)` instead.

Explain the changed result using the matching lesson.

What happens visibly if the caller stores needed but omits print(needed)?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Give the recommender separate validation, eligibility and scoring functions whose inputs and results can be tested independently.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Give the recommender separate validation, eligibility and scoring functions whose inputs and results can be tested independently.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional library computes total spine labels needed for several equal shelves.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? needed receives 24; print(needed) displays 24.

1. Function body
2. Caller prints
3. Call with arguments
4. Return value

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Caller prints needed receives 24;

`print(needed)` displays 24. A function separates one named job from the surrounding program. Arguments supply data for this call; `return` supplies its result. A `print` statement in the caller is a different action. Clear parameter names help detect swapping or misusing inputs without assuming every function prints automatically.

Practice 2: explain the deciding evidence

Changed-case practice: Call `labels(4,0)` instead. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. The returned result is 0, and the caller prints 0.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The returned result is 0, and the caller prints 0. Four shelves times zero labels per shelf is zero.

Repair a missing building block

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script.

Short practice: If `print(total)` moves before `total=total+2`, what gets printed for the original input?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
